

Midterm

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```
#remove.packages("h2o")
#install.packages("h2o")
```

Prerequisites

```
# Helper packages
library(rsample) # for creating our train-test splits

## Warning: package 'rsample' was built under R version 4.3.2
library(recipes) # for minor feature engineering tasks
```

```
## Warning: package 'recipes' was built under R version 4.3.3
## Loading required package: dplyr
##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##   filter, lag
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
##
## Attaching package: 'recipes'
## The following object is masked from 'package:stats':
##
##   step
```

```
library(dplyr) # for data manipulation
library(ggplot2) # for data visualization
library(tidyr) # for data reshaping

# Modeling packages
library(h2o) # for fitting stacked models
```

```
##
## -----
##
## Your next step is to start H2O:
##   > h2o.init()
```

```

##
## For H2O package documentation, ask for help:
##   > ??h2o
##
## After starting H2O, you can use the Web UI at http://localhost:54321
## For more information visit https://docs.h2o.ai
##
## -----
##
## Attaching package: 'h2o'
##
## The following objects are masked from 'package:stats':
##
##   cor, sd, var
##
## The following objects are masked from 'package:base':
##
##   &&, %*%, %in%, ||, apply, as.factor, as.numeric, colnames,
##   colnames<-, ifelse, is.character, is.factor, is.numeric, log,
##   log10, log1p, log2, round, signif, trunc
library(recipes)  # for ML recipes
library(xgboost)  # for fitting GBMs

## Warning: package 'xgboost' was built under R version 4.3.3
##
## Attaching package: 'xgboost'
##
## The following object is masked from 'package:dplyr':
##
##   slice
h2o.no_progress() # turn off progress bars
h2o.init(max_mem_size = "5g") # connect to H2O instance

## Connection successful!
##
## R is connected to the H2O cluster:
##   H2O cluster uptime:      6 hours 34 minutes
##   H2O cluster timezone:    America/Los_Angeles
##   H2O data parsing timezone: UTC
##   H2O cluster version:    3.44.0.3
##   H2O cluster version age: 1 year, 4 months and 11 days
##   H2O cluster name:       H2O_started_from_R_tifle_nit536
##   H2O cluster total nodes: 1
##   H2O cluster total memory: 1.75 GB
##   H2O cluster total cores: 8
##   H2O cluster allowed cores: 8
##   H2O cluster healthy:    TRUE
##   H2O Connection ip:      localhost
##   H2O Connection port:    54321
##   H2O Connection proxy:   NA
##   H2O Internal Security:  FALSE
##   R Version:              R version 4.3.1 (2023-06-16)
##
## Warning in h2o.clusterInfo():

```

Your H2O cluster version is (1 year, 4 months and 11 days) old. There may be a newer version available.
Please download and install the latest version from: <https://h2o-release.s3.amazonaws.com/h2o/latest>.

Ames Housing Dataset

```
# Load and split the Ames housing data
ames <- AmesHousing::make_ames()
set.seed(123) # for reproducibility
split <- initial_split(ames, strata = "Sale_Price")
ames_train <- training(split)
ames_test <- testing(split)

# Make sure we have consistent categorical levels
blueprint <- recipe(Sale_Price ~ ., data = ames_train) %>%
  step_other(all_nominal(), threshold = 0.005)

# Create training & test sets for h2o
train_h2o <- prep(blueprint, training = ames_train, retain = TRUE) %>%
  juice() %>%
  as.h2o()
test_h2o <- prep(blueprint, training = ames_train) %>%
  bake(new_data = ames_test) %>%
  as.h2o()

# Get response and feature names
Y <- "Sale_Price"
X <- setdiff(names(ames_train), Y)
```

Base Learner - Models

Regularized Regression Base Learner

Random Forest Base Learner

```
# Train & cross-validate a RF model w/ Random Search
rf_params <- list(
  ntrees = c(100, 200, 300),
  max_depth = c(10, 15, 20),
  sample_rate = c(0.7, 0.8, 0.9)
)
grid_id_rf <- paste0("rf_grid", as.integer(Sys.time()))
rf_search <- h2o.grid(
  algorithm = "randomForest",
  grid_id = grid_id_rf,
  x = X, y = Y,
  training_frame = train_h2o,
  hyper_params = rf_params,
  search_criteria = list(strategy = "RandomDiscrete", max_models = 5, seed = 123),
  nfolds = 2,
  fold_assignment = "Modulo",
  keep_cross_validation_predictions = TRUE,
  stopping_rounds = 10,
  stopping_metric = "RMSE",
  stopping_tolerance = 0,
```

```

    seed = 123
)

best_rf <- h2o.getGrid("rf_grid", sort_by = "rmse", decreasing = FALSE)@model_ids[[1]]

```

Gradient Boosting Base Learner

```

# Train & cross-validate a GBM model w/ Random Search
gbm_params <- list(
  learn_rate = c(0.01, 0.05),
  max_depth = c(5, 7, 9),
  sample_rate = c(0.7, 0.8, 0.9)
)

grid_id_gbm <- paste0("gbm_grid_", as.integer(Sys.time()))

gbm_search <- h2o.grid(
  algorithm = "gbm",
  grid_id = grid_id_gbm,
  x = X, y = Y,
  training_frame = train_h2o,
  hyper_params = gbm_params,
  search_criteria = list(strategy = "RandomDiscrete", max_models = 5, seed = 123),
  nfolds = 2,
  fold_assignment = "Modulo",
  keep_cross_validation_predictions = TRUE,
  stopping_rounds = 10,
  stopping_metric = "RMSE",
  stopping_tolerance = 0,
  seed = 123
)

best_gbm <- h2o.getGrid("gbm_grid", sort_by = "rmse", decreasing = FALSE)@model_ids[[1]]

```

XGBoost Base Learner

```

# Train & cross-validate an XGBoost model w/ Random Search
xgb_params <- list(
  learn_rate = c(0.01, 0.05, 0.1),
  max_depth = c(3, 5, 7),
  sample_rate = c(0.7, 0.8, 0.9)
)

grid_id_xgb <- paste0("xgb_grid_", as.integer(Sys.time()))

xgb_search <- h2o.grid(
  algorithm = "xgboost",
  grid_id = grid_id_xgb,
  x = X, y = Y,
  training_frame = train_h2o,
  hyper_params = xgb_params,
  search_criteria = list(strategy = "RandomDiscrete", max_models = 5, seed = 123),
  categorical_encoding = "LabelEncoder",
  nfolds = 2,
  fold_assignment = "Modulo",

```

```

keep_cross_validation_predictions = TRUE,
stopping_rounds = 10,
stopping_metric = "RMSE",
stopping_tolerance = 0,
seed = 123
)

best_xgb <- h2o.getGrid("xgb_grid", sort_by = "rmse", decreasing = FALSE)@model_ids[[1]]

```

Neural Network Stack Model (Super Learner)

Results

```

# Load actual H2O model objects from their IDs
model_ids <- list("glm_grid_model_3", "rf_grid_model_1", "gbm_grid_model_12", "xgb_grid_model_5")
models <- purrr::map(model_ids, h2o.getModel)

# Now define get_rmse function and apply it
get_rmse <- function(model) {
  perf <- h2o.performance(model, newdata = test_h2o)
  h2o.rmse(perf)
}

# Get RMSEs
rmse_values <- purrr::map_dbl(models, get_rmse)

rmse_values

## [1] 26204.91 22231.09 21003.10 23170.78

# Stacked results
h2o.performance(ensemble_nn, newdata = test_h2o)@metrics$RMSE

## [1] 21425.67

get_cv_preds <- function(model) {
  frame_id <- model@model$cross_validation_holdout_predictions_frame_id$name
  if (!is.null(frame_id)) {
    return(as.vector(h2o.getFrame(as.character(frame_id))))
  } else {
    warning(paste("No CV predictions found for model:", model@model_id))
    return(rep(NA, model@parameters$training_frame$num_rows))
  }
}

# Load models by ID
best_glm <- h2o.getModel("glm_grid_model_3")
best_rf <- h2o.getModel("rf_grid_model_1")
best_gbm <- h2o.getModel("gbm_grid_model_12")
best_xgb <- h2o.getModel("xgb_grid_model_5")
best_nn <- h2o.getModel("nn_ensemble_randomsearch") # replace if needed

# Collect predictions
pred_df <- data.frame(
  GLM_pred = get_cv_preds(best_glm),

```

```

RF_pred = get_cv_preds(best_rf),
GBM_pred = get_cv_preds(best_gbm),
XGB_pred = get_cv_preds(best_xgb)
)

# Drop rows with NA (in case any model didn't return predictions)
pred_df <- na.omit(pred_df)

# Correlation matrix
cor(pred_df)

```

```

##           GLM_pred  RF_pred  GBM_pred  XGB_pred
## GLM_pred  1.0000000  0.9433703  0.9508229  0.9293876
## RF_pred   0.9433703  1.0000000  0.9868962  0.9788307
## GBM_pred  0.9508229  0.9868962  1.0000000  0.9737024
## XGB_pred  0.9293876  0.9788307  0.9737024  1.0000000

```

Visualization / Comparison of Models

```

# Create a data frame with all model results
model_comparison <- data.frame(
  Model = c("GLM", "Random Forest", "GBM", "XGBoost", "Neural Network Ensemble"),
  RMSE = c(
    get_rmse(best_glm),
    get_rmse(best_rf),
    get_rmse(best_gbm),
    get_rmse(best_xgb),
    h2o.performance(ensemble_nn, newdata = test_h2o)$metrics$RMSE
  )
)

# Sort models by performance (lowest RMSE first)
model_comparison <- model_comparison[order(model_comparison$RMSE), ]

# Show Model Comparison
model_comparison

```

```

##           Model      RMSE
## 3           GBM 21003.10
## 5 Neural Network Ensemble 21425.67
## 2           Random Forest 22231.09
## 4           XGBoost 23170.78
## 1           GLM 26204.91

```

```

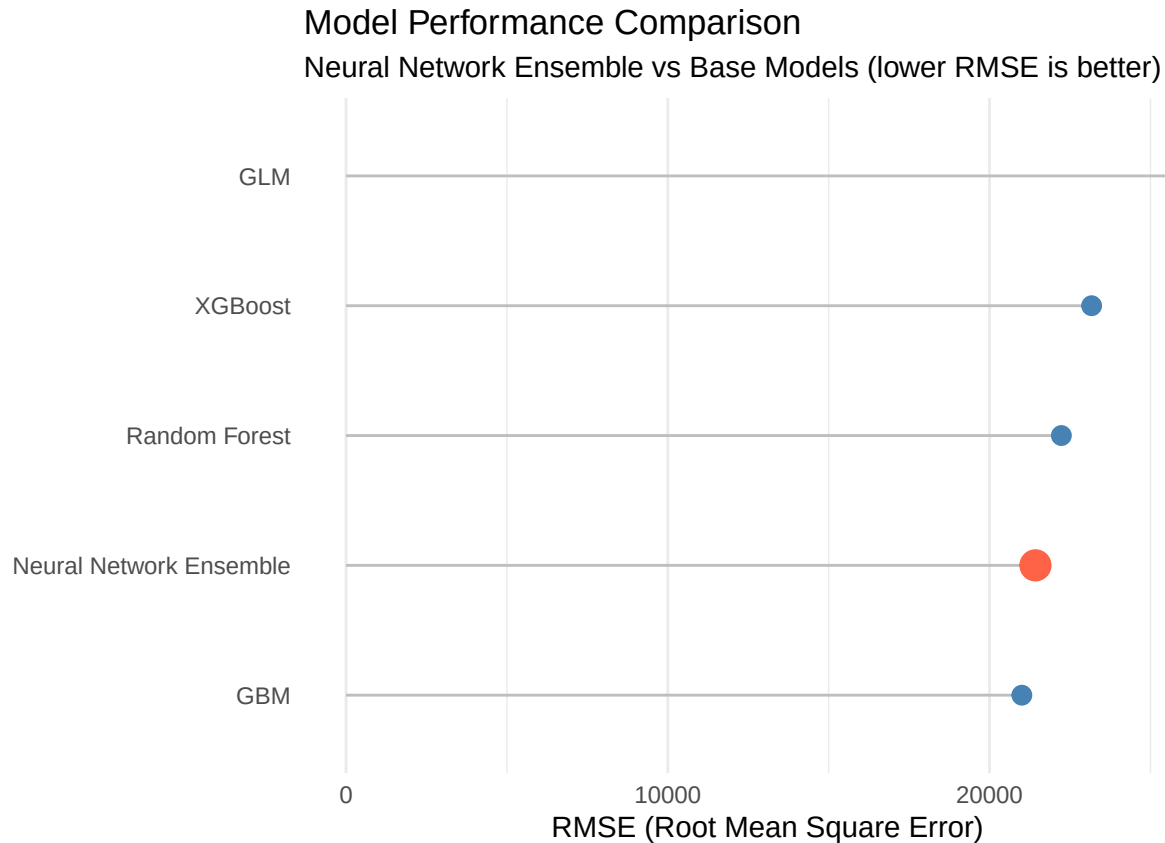
# Compare Ensemble to Others
ggplot(model_comparison, aes(x = reorder(Model, RMSE))) +
  geom_segment(aes(xend = Model, y = 0, yend = RMSE), color = "gray") +
  geom_point(aes(y = RMSE, color = Model == "Neural Network Ensemble",
    size = Model == "Neural Network Ensemble")) +
  scale_color_manual(values = c("steelblue", "tomato"), guide = "none") +
  scale_size_manual(values = c(3, 5), guide = "none") +
  coord_flip() +
  theme_minimal() +
  labs(title = "Model Performance Comparison",

```

```

subtitle = "Neural Network Ensemble vs Base Models (lower RMSE is better)",
y = "RMSE (Root Mean Square Error)",
x = NULL) +
theme(panel.grid.major.y = element_blank())

```



```

# Calculate percentage improvement over worst model
baseline_rmse <- max(model_comparison$RMSE)
model_comparison$Improvement <- (baseline_rmse - model_comparison$RMSE) / baseline_rmse * 100

# Add percentage improvement labels to the comparison
clean_chart <- ggplot(model_comparison, aes(x = reorder(Model, -RMSE), y = RMSE)) +
  geom_bar(stat = "identity", width = 0.7,
    fill = ifelse(model_comparison$Model == "NN_Ensemble", "tomato", "steelblue")) +
  # Position labels ABOVE the bars with proper spacing
  geom_text(aes(label = format(round(RMSE, 2), big.mark=","),
    y = RMSE + 1000), # Position above bars with 1000 units of space
    vjust = 0,
    size = 3.5) +
  # Add smaller improvement text below first text
  geom_text(aes(label = paste0("(", round(Improvement, 1), "% better)"),
    y = RMSE + 1000),
    vjust = -1.5,
    size = 3) +
  theme_classic() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
    plot.title = element_text(hjust = 0.5),

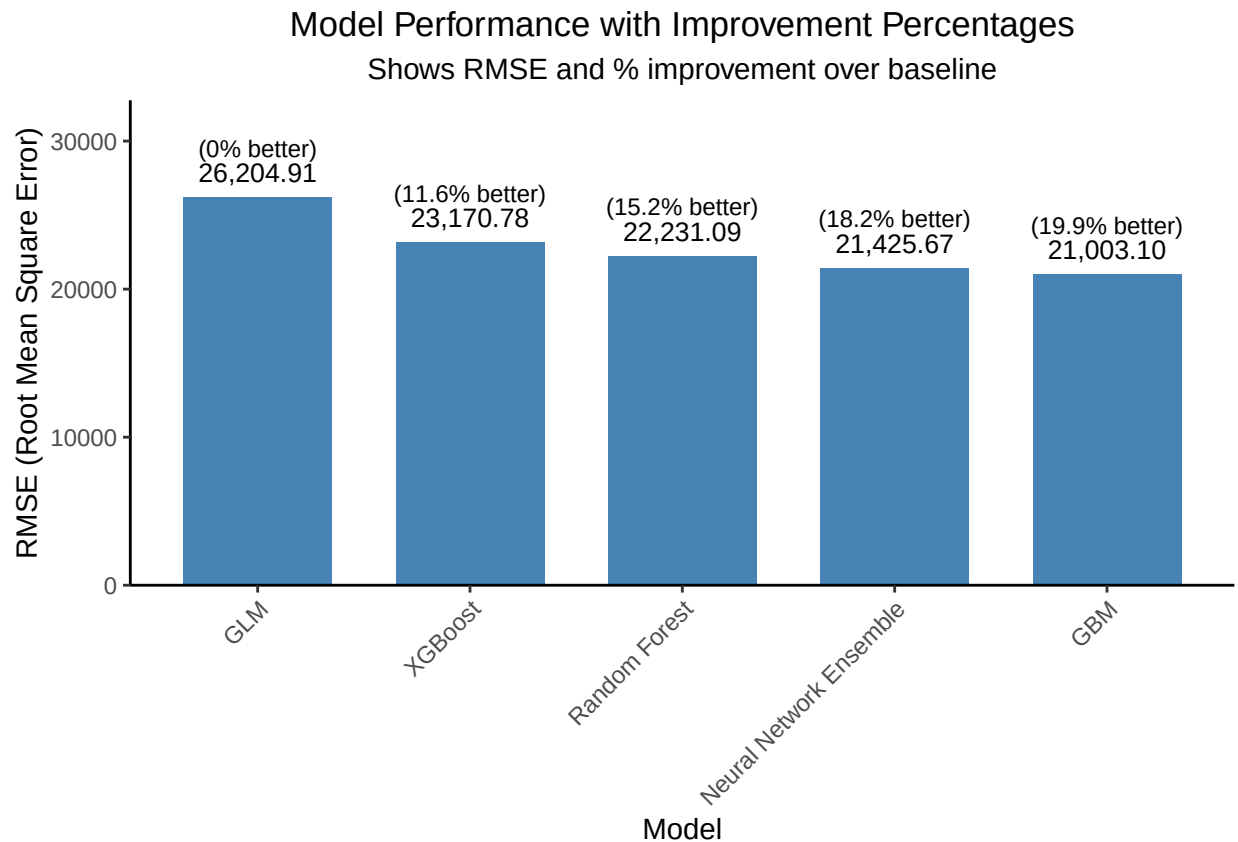
```

```

    plot.subtitle = element_text(hjust = 0.5)) +
  labs(title = "Model Performance with Improvement Percentages",
       subtitle = "Shows RMSE and % improvement over baseline",
       x = "Model",
       y = "RMSE (Root Mean Square Error)") +
  scale_y_continuous(limits = c(0, max(model_comparison$RMSE) * 1.25), # Add 25% space at top
                    expand = c(0, 0)) # Remove default expansion

print(clean_chart)

```



```

# Function to get RMSE from H2O models
get_rmse <- function(model) {
  results <- h2o.performance(model, newdata = test_h2o)
  results@metrics$RMSE
}

# Get RMSE for all models
rmse_values <- c(
  GLM = get_rmse(best_glm),
  RF = get_rmse(best_rf),
  GBM = get_rmse(best_gbm),
  XGBoost = get_rmse(best_xgb),
  NN_Ensemble = h2o.performance(ensemble_nn, newdata = test_h2o)@metrics$RMSE
)

# Create a data frame with all model results

```



```
model_comparison <- data.frame(
  Model = names(rmse_values),
  RMSE = unname(rmse_values)
)
```

Comparison Table

```
# Sort models by performance (lowest RMSE first)
model_comparison <- model_comparison[order(model_comparison$RMSE), ]

# Calculate percentage improvement over worst model (highest RMSE)
baseline_rmse <- max(model_comparison$RMSE)
model_comparison$Improvement <- round((baseline_rmse - model_comparison$RMSE) / baseline_rmse * 100, 2)

# Calculate percentage improvement over best base model
best_base_rmse <- min(model_comparison$RMSE[model_comparison$Model != "NN_Ensemble"])
model_comparison$Improvement_Over_Best_Base <- NA
model_comparison$Improvement_Over_Best_Base[model_comparison$Model == "NN_Ensemble"] <-
  round((best_base_rmse - model_comparison$RMSE[model_comparison$Model == "NN_Ensemble"]) / best_base_rmse * 100, 2)

# Create a nicely formatted performance comparison table
print("Model Performance Comparison (lower RMSE is better)")

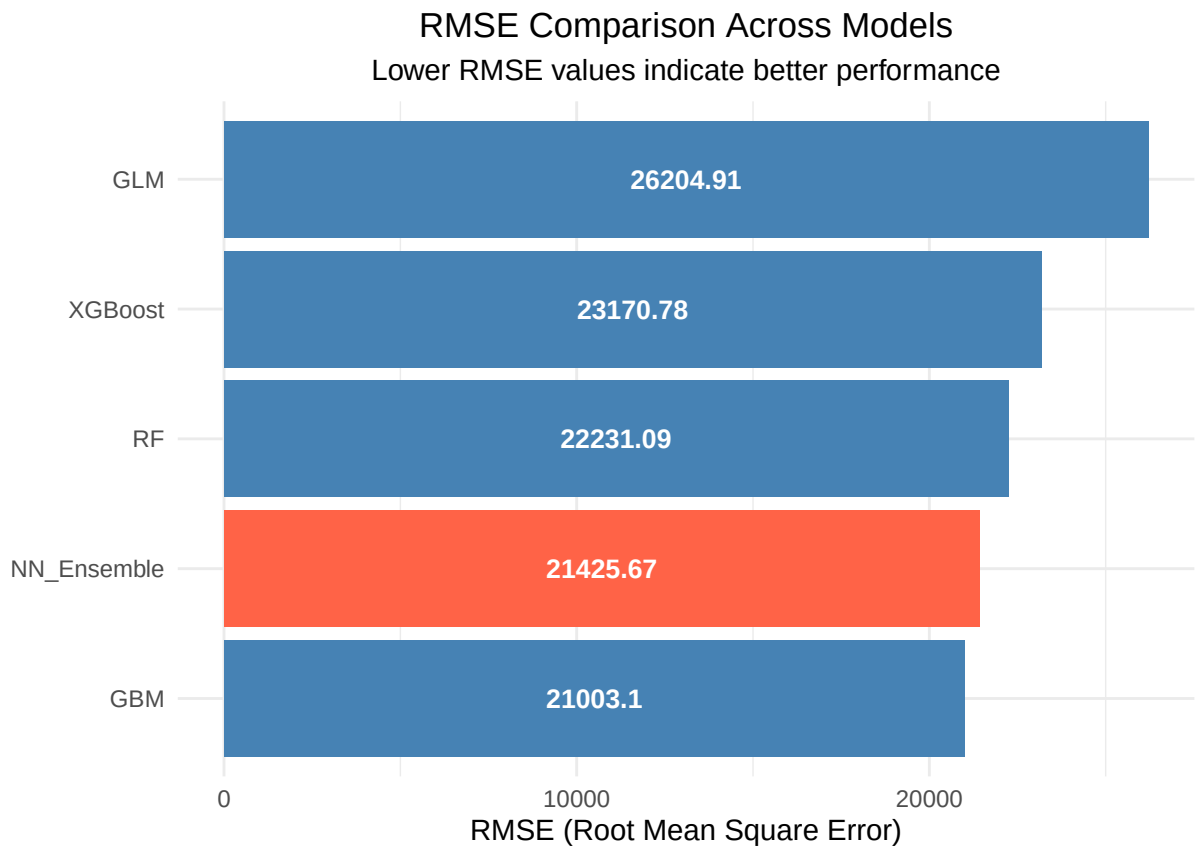
## [1] "Model Performance Comparison (lower RMSE is better)"
print(model_comparison)
```

	Model	RMSE	Improvement	Improvement_Over_Best_Base
## 3	GBM	21003.10	19.85	NA
## 5	NN_Ensemble	21425.67	18.24	-2.01
## 2	RF	22231.09	15.16	NA
## 4	XGBoost	23170.78	11.58	NA
## 1	GLM	26204.91	0.00	NA

Comparison via Bar Chart (Red for Stacked NN)

```
# Basic bar chart of RMSE values
bar_chart <- ggplot(model_comparison, aes(x = reorder(Model, RMSE), y = RMSE)) +
  geom_bar(stat = "identity",
    aes(fill = Model == "NN_Ensemble"),
    show.legend = FALSE) +
  scale_fill_manual(values = c("steelblue", "tomato")) +
  geom_text(aes(label = round(RMSE, 2)),
    position = position_stack(vjust = 0.5), # Position text in middle of bars
    color = "white", # White text for better contrast
    fontface = "bold", size = 3.5) + # Bold text
  coord_flip() + # Flip for horizontal bars
  theme_minimal() +
  labs(title = "RMSE Comparison Across Models",
    subtitle = "Lower RMSE values indicate better performance",
    x = "",
    y = "RMSE (Root Mean Square Error)") +
  theme(plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5))
```

bar_chart



Summary Statistics

```
# Identify best model
best_model <- model_comparison$Model[1]
best_rmse <- model_comparison$RMSE[1]

# Find the best base model (not ensemble)
best_base_model <- model_comparison$Model[model_comparison$Model != "NN_Ensemble"][1]
best_base_rmse <- model_comparison$RMSE[model_comparison$Model == best_base_model]

# Calculate improvement of ensemble over best base model
if (best_model == "NN_Ensemble") {
  ensemble_improvement <- ((best_base_rmse - best_rmse) / best_base_rmse) * 100
  ensemble_status <- paste("The Neural Network Ensemble improved upon the best base model by",
    round(ensemble_improvement, 2), "%")
} else {
  ensemble_status <- "The Neural Network Ensemble did not outperform the best base model"
}

# Print summary
cat("\n===== SUMMARY =====\n")
```

```
##
## ===== SUMMARY =====
```

```

cat("Best overall model:", best_model, "with RMSE =", round(best_rmse, 3), "\n")

## Best overall model: GBM with RMSE = 21003.1
cat("Best base model:", best_base_model, "with RMSE =", round(best_base_rmse, 3), "\n")

## Best base model: GBM with RMSE = 21003.1
cat(ensemble_status, "\n")

## The Neural Network Ensemble did not outperform the best base model
cat("=====\n")

## =====

```